

Expanding and factorizing expressions

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Quadratic expressions

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THE SAME EXPRESSION CAN BE WRITTEN IN DIFFERENT WAYS—MULTIPLIED OUT (EXPANDED) OR GROUPED INTO ITS COMMON FACTORS (FACTORIZED).

How to expand an expression

The same expression can be written in a variety of ways, depending on how it will be used. Expanding an expression involves multiplying all the parts it contains (terms) and writing it out in full.

$$4(a + 3) = 4 \times a + 4 \times 3 = 4a + 12$$

number outside is multiplied by each number inside

first term is multiplied by number

second term is multiplied by number

$4 \times a = 4a$

$4 \times 3 = 12$

sign between terms remains the same

To expand an expression with a number outside a parenthesis, multiply all the terms inside the parenthesis by that number.

Multiply each term inside the parenthesis by the number outside. The sign between the two terms (letters and numbers) remains the same.

Simplify the resulting terms to show the expanded expression in its final form. Here, $4 \times a$ is simplified to $4a$ and 4×3 to 12.

Expanding multiple parentheses

To expand an expression that contains two parentheses, each part of the first one is multiplied by each part of the second parenthesis. To do this, split up the first (blue) parenthesis into its parts. Multiply the second (yellow) parenthesis by the first part and then by the second part of the first parenthesis.

$$(3x + 1)(2y + 3) = 3x(2y + 3) + 1(2y + 3) = 6xy + 9x + 2y + 3$$

second parenthesis multiplied by first part of first parenthesis

second parenthesis multiplied by second term of first parenthesis

first parenthesis

second parenthesis

$3x \times 2y = 6xy$

$3x \times 3 = 9x$

$1 \times 2y = 2y$

$1 \times 3 = 3$

these signs remain

To expand an expression of two parentheses, multiply all the terms of the second by all the terms of the first.

Break down the first parenthesis into its terms. Multiply the second parenthesis by each term from the first in turn.

Simplify the resulting terms by carrying out each multiplication. The signs remain the same.

Squaring a parenthesis

Squaring a parenthesis simply means multiplying a parenthesis by itself. Write it out as two parentheses next to each other, and then multiply it to expand as shown above.

$$(x - 3)^2 = (x - 3)(x - 3) = x(x - 3) - 3(x - 3) = x^2 - 3x - 3x + 9 = x^2 - 6x + 9$$

parenthesis means multiply

multiply second parenthesis by first part of first one

sign remains the same

multiply second parenthesis by second part of first parenthesis

$x \times x = x^2$

$x \times -3 = -3x$

$-3 \times x = -3x$

$-3 \times -3 = 9$

To expand a squared parenthesis, first write the expression out as two parentheses next to each other.

Split the first parenthesis into its terms and multiply the second parenthesis by each term in turn.

Simplify the resulting terms, making sure to multiply their signs correctly. Finally, add or subtract like terms (see pp.172–173) together.